

RAY

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1 PROJECT URL

<https://www.zhangweidi.com/ray>



2 DESCRIPTION OF ARTWORK

In 1922, Man Ray, a significant contributor to Dada and Surrealism movements, invented the term “Rayograph” to describe his cameraless darkroom created photo images. He revealed the dreamlike characteristics of daily objects by placing them in photosensitized paper and exposing them to light. His Rayographs are situated in-between visual representation and abstraction, pointing to a new cameraless photographic method. Man Ray wrote in 1921 that he wanted to “make my photography automatic—to use my camera as I would a typewriter.”

A Rayograph (photogram) is an image created by exposing light-sensitive photographic paper to light, where objects are placed on it to block areas of the paper. Variations of this technique are also implemented for scientific purposes such as X-Rays, which are made by using external radiation to produce images of our body, and other internal structures for diagnostic purposes. Today, We live under surveillance, our life moments and bodies have been captured, evaluated, and judged by the intelligent system behind the cameras. Image-making practices are beyond merely taking

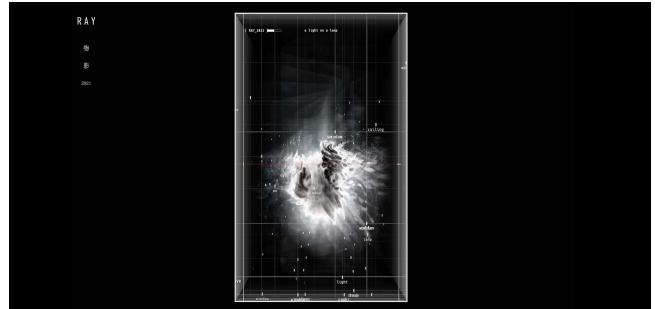
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pictures that record light, such as operative images and generative images that emphasize instructional practices and autonomous systems. Countless images are no longer visual representations of the world made for humans to perceive but programmable data-based visualizations which result through automatic operations. The notion of the cameraless image-making process shifted from Rayograph to the rule-based activities and synchronic data-to-data relationships.



RAY provides a conceptual response to the changing notion of images and cameraless image-making technique under the gazes of machines through re-interpreting Rayograph (photogram). RAY is an intelligent, responsive system that converts the live-streaming image of audiences into a responsive visualization of new Rayograph (photogram) through learning over 3000 image pairs of human portraits and Rayograph. The intelligent system RAY consists of two parts: 1. generic image-to-image translation using conditional adversarial networks (GANs in the conditional setting). A neural network is trained to translate features from human portraits to a database of collected photograms of objects. 2. A computer vision system is used to localize and describe salient regions in images in natural language. Descriptive sentences are generated in real-time to describe the participants captured by the camera in the installation. The artistic goal of visualization is to associate generative visualization with the modern cameraless image-making process by emphasizing the action of light through revealing and concealing information. Thus, RAY further develops and visualizes the machine-generated image-data algorithmically within the tradition of light painting – a technique of taking a long exposure of a

moving light source – to create the moving composition. Inspired by the Advanced Image Technology (AIT) body scanner used in the airport that reveals information of potential threat, the design of RAY's interface presents layers of information as a scanner interface, including virtual sculpture (generative visualization), textual elements (descriptive words), a 3D frame, and indicators. In the installation, a camera is hidden in the space to capture audience presence and send the live streaming feed to the system RAY, which results in an ever-changing interactive experience.

3 ARTIST INFORMATION

Weidi Zhang is an LA-based new media artist/researcher/designer. Her current research and media art practices investigate the image

data-based multimedia assemblage in the context of experimental data visualization, Interactive Intelligence system design, and immersive media. Her works are featured at different international venues, such as the SIGGRAPH, SIGGRAPH ASIA, ISEA, Times Art Museum, IEEE VISAP, SwissNex Gallery, Japan Media Art Festival, Lumen Prize, and others. Her immersive audiovisual is screened in the international Fulldome Festivals including Metaxis Festival Planetarium 1 (Russia), Zeiss-Planetarium (Germany), and others. Currently, she is a Ph.D. candidate in the Media Arts and Technology Program, UCSB.